

Project MediLong

Data Pipelines Squad — Detailed Report

OCR/NLP + Ultrasound CV Stub · 27-Day Sprint · Sept 2 – Sept 28

Squad	Data Pipelines (merged OCR/NLP + CV)
Members	Mani, Keerthi, Kamalesh C
Sprint Window	Sept 2 – Sept 28, 2026 (27-day sprint)

Mission

Turn unstructured uploads — lab PDFs and ultrasound images — into clean, contract-shaped structured data. This squad covers two of the three pillars; suggested internal split: Mani + Keerthi lead OCR/NLP, Kamalesh C leads the CV stub with support as needed.

Contract Slice Owned

pillar_2_ultrasound (mode, follicle_count, volume_ml, confidence) and pillar_3_labs (all 10 biomarkers + source_confidence).

Sprint-by-Sprint Tasks

Sprint 1 (Sept 2–8)

- OCR/NLP: collect real/synthetic lab PDFs across 3+ lab-chain formats; define a field-level extraction accuracy metric.
- CV stub: confirm output format with Backend (mode: "stub"); build image validation (file type, size, corruption checks) — this part is real, not mocked.
- CV stub: source the Kaggle PCOS ultrasound dataset in case a coarse pretrained detector gets wired in later.

Sprint 2 (Sept 9–15)

- OCR/NLP: robust table/field parsing for all 10 biomarkers; null + confidence score on missing/noisy fields rather than a crash.
- CV stub: finalize stub logic returning plausible follicle counts/volumes within clinical ranges; label mode: "stub" on every response.
- Stretch goal if time allows: wire an off-the-shelf pretrained detector for a coarse real follicle count instead of pure randomization.

Sprint 3 (Sept 16–22)

- Both pipelines go live behind Backend's /upload/* and /predict orchestration.
- Joint testing with Backend on real PDF/image samples, not synthetic fixtures.

Sprint 4 (Sept 23–28)

- Edge cases: non-PCOS/irrelevant lab reports, corrupted images, partially missing biomarker panels.
- Confirm every response degrades gracefully (confidence score drop, never a hard failure).

Dependencies

- Blocked by: Backend's /upload/* endpoint contract.
- Blocks: ML Fusion (needs real biomarker + stub CV output to train/validate against), Frontend (review screen depends on source_confidence).

Definition of Done

- All 10 lab biomarkers extracted with confidence scores on real-world PDF samples.
- CV stub schema-correct and clearly flagged as non-diagnostic in every response.
- No unhandled crashes on malformed input.